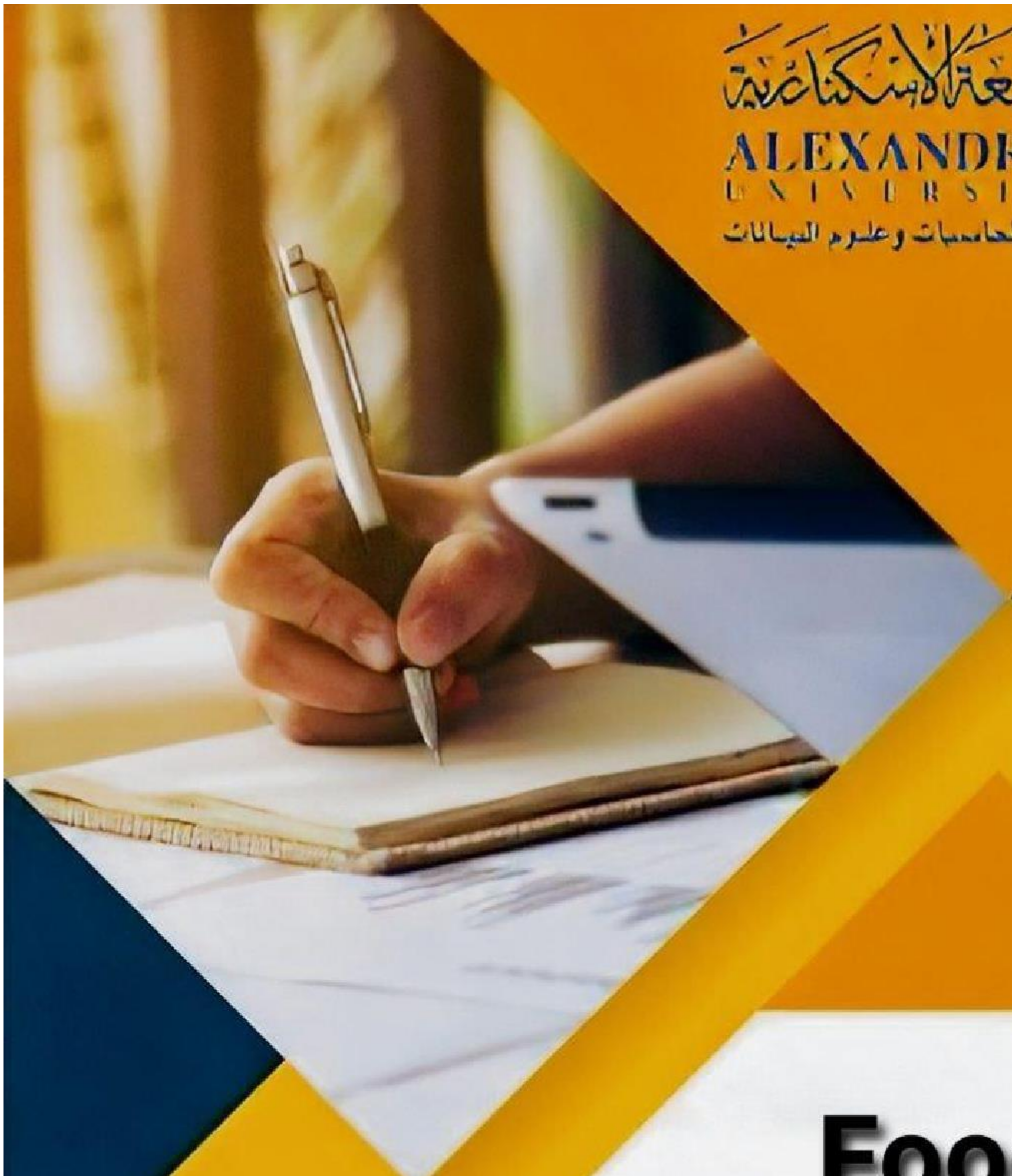

جامعة الإسكندرية
ALEXANDRIA
UNIVERSITY
كلية الحاسبات وعلوم البيانات

Food Preferences

Project Report

Survey Methodology

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[1] Content: -

[2] Introduction: -

This report is established to summarize the team's progress throughout the project.

First of all, let me introduce out Topic. Our Survey is about food preferences. Food preferences are an essential aspects of people's daily lives, impacting their health, social interactions and cultural experiences. As a result, understanding people's food preferences is crucial for improving food services and marketing strategies.

[3] Phase 1: Proposal About our Survey: -

To begin with, we explained why this survey is important as we needed the data that has been collected to develop new products and menu items that cater to people's preference and dietary needs and reduce food wastes.

We have introduced **seven** important points that we had to take in consideration throughout the project which are: -

1) Objectives

- Gather data on people's food preferences in term of their favorite cuisines, ingredients and dietary restrictions.
- Determine the most preferred food types and explore how they vary from one person to another.
- Develop a suitable menu for a restaurant near a university.

2) Participants

- The survey was conducted on a numbers of students at university ranging from 400 to 500 students using an anonymous online platform, which is accessible to a large number of participants.

3) Survey Design

- The survey was designed in the form of an online questionnaire of 20 – 30 questions providing information about food choices, habits, restrictions and some demographic questions.

4) Data Analysis

- Where we included descriptive analysis, correlation analysis, regression analysis and data visualization.

5) Timeline

- We took about 8 weeks to finish the whole project and delivered it on time.

6) Budgets

- We have carefully planned the budget for each stage of the survey and ensured that we had sufficient resources to complete our project.

7) Ethics and confidentiality

- By including this section, we were able to ensure that the participant's privacy and rights were protected throughout the survey.

[4] *Phase 2: The Questionnaire: -*

In the second step, we studied the proposal well and understood the objectives so that we can construct our questionnaire in an efficient way.

The questionnaire was designed to be easy to complete and it took no more than 10 – 15 minutes to finish. We have designed it to have 30 questions, 7 opened questions and 23 multiple choice questions.

Questions were constructed to be easily understood. We used multiple choices for most of the questions with an option for respondents to select more than one answer if applicable.

The questionnaire consists of demographic questions and main survey questions. We also had a question to be repeated to detect whether the response is filled randomly or not so that we reduce the biasness as little as possible.

We constructed a lot of questions and then filtered them and chose the most suitable questions to be embedded in the questionnaire.

[5] Phase 3: Sampling and Pretests: -

After constructing our questionnaire, we started to distribute the questionnaire with some feedback questions on a sample of our population and gather their feedbacks to know if there is any problem with the questions in the survey.

The sample size -we tested our questionnaire on- was 107.Then we started analyzing their feedbacks to know what updates we need to do on the questionnaire. Some feedbacks were limited into some modifications which we did like:

- Some questions needed to have more variety of options.
- Others need to have an option ‘none’ in the answer.

After that, Feedbacks were sent back to phase 2 to modify the questions and get the questionnaire well done from all aspects.

[6] Phase 4: Data Collection & pre-processing: -

Our questionnaire is now ready to be delivered to our population. In this phase, we started gathering data from respondents to analyze it and gain some insights that would enhance our decisions and help us develop the best solution.

We collected 527 responses from the population and started preparing data for analysis by using python. First, we started translating Arabic responses into English so that all data would be similar and easier to be analyzed. Then, we checked if there is missing values in the dataset.

```
df.isnull().sum()
```

```
13]
0                                                                       طابع زمني
0                                                                        النتيجة
?What's your Name                                                              6
 Which of the following is your favourite cuisine                              0
?What's your favourite chef                                                  132
?How do you define your food regime                                            0
?How often do you go shopping for food                                        34
?How would you describe your diet                                              0
   İs your country famous for any foods ? What are they                       73
What meal do you prefer eating through day ?                                   0
 Are you allergic to any of the following food items?(please select all that apply)    137
?Which do you prefer for breakfast                                             0
 Choose What kind of food you consume regularly for lunch                      0
How much do you prefer salt in your food?                                      3
?Do you like spicy food                                                        0
?What do you prefer for dinner                                                 2
Do you eat various types of vegetables or fruit?                               7
If you eat vegetables or fruits mention what you prefer?                       3
 Please mention which ingredients that you dislike in meals                  112
? How often do you consume fast food                                           5
What kind of fast food do you usually consume?                                 0
Please select your favourite restaurant                                        3
?How do you like your food                                                     0
Describe  reasons why people enjoy fast food                                   7
Which dessert of the following you prefer?                                     0
...
?What is your gender                                                           0
?What's your age                                                               0
?What's your education                                                         0
?How much is your monthly income                                              34
dtype: int64
```

We found out that we would lose a large amount of our data if we decided to remove all records that contains null values, so we decided to replace NAN values with the most frequent occurred item in each column.

```python
df = pd.read_excel('C:\\Users\\User\\Desktop\\responses.xlsx')
# numeric columns
for col in df.select_dtypes(include=['number']):
    df[col].fillna(df[col].mean(), inplace=True)

# categorical columns
for col in df.select_dtypes(include=['object']):
    df[col].fillna(df[col].mode()[0], inplace=True)

print(df)
```

```
                    النتيجة  طابع زمني ?What's your Name  \
0   2023-04-27 20:54:42.391        0     Salma Soliman
1   2023-05-01 18:07:54.665        0   Aya Ahmed Rehan
2   2023-05-01 18:19:54.363        0            Mariam
3   2023-05-01 19:13:09.569        0    Tasneem Rehan
4   2023-05-01 20:51:19.904        0      Shahd asaya
..                      ...      ...               ...
521 2023-05-07 20:54:58.056        0      Ahmed desoky
522 2023-05-07 21:02:05.498        0         The big M
523 2023-05-07 21:14:07.762        0           NOURHAN
524 2023-05-07 21:38:44.041        0         Mona Badr
525 2023-05-07 22:35:10.737        0      Yasser Fouad

     Which of the following is your favourite cuisine   \
0                                        Egyptian food
1                                        Egyptian food
2                         Italian food, Egyptian food
3                                        Egyptian food
4                                        Egyptian food
..                                                 ...
521                                      Egyptian food
522          Italian food, Egyptian food, Lebanese food
523                                       Italian food
524                                      Egyptian food
525                                       Chinese food
```

In the second step, we started removing inconsistent data by testing if the age that the respondent has entered fits with education level or not.

```python
df["Age"]=df["Age"].str.replace("[YearsYearyearsoldN?MatiamyaerGoodPeter]","",
regex=False)
```

```python
df['Age'] = pd.to_numeric(df['Age'], errors='coerce')

df['Age'].fillna(df['Age'].mean() , inplace=True)
```

```python
for i in range(len(df)):
    if df.loc[i,"Age"] < 18:
        df.loc[i,"Age"] = df['Age'].mean()
```

```
df['Age'] = df['Age'].astype(int)
```

```
for i in range(len(df)):

    if df.loc[i,"Education"] == "Teaching assistant" and
      (df.loc[i,"Age"] <23 or df.loc[i,"Age"] > 30):
        df.loc[i,"Education"] =None

    elif df.loc[i,"Education"] == "Doctor" and df.loc[i,"Age"] < 25:
        df.loc[i,"Education"] =None

    elif df.loc[i,"Education"] == "Level 2" and df.loc[i,"Age"] < 19:
        df.loc[i,"Education"] = None
```

```

    elif df.loc[i,"Education"] == "Level 3" and df.loc[i,"Age"] < 20:
        df.loc[i,"Education"] = None

    elif df.loc[i,"Education"] == "Level 4" and df.loc[i,"Age"] < 21:
        df.loc[i,"Education"] = None

    elif df.loc[i,"Education"] == "Level 5" and df.loc[i,"Age"] < 22:
        df.loc[i,"Education"] = None

    elif df.loc[i,"Education"] == "Level 6" and df.loc[i,"Age"] < 23:
        df.loc[i,"Education"] =None
    else:
        df.loc[i,"Education"] = df.loc[i,"Education"]
```

```
df.dropna(subset=["Education"], inplace=True)
```

At last, we checked if data was entered randomly in the questionnaire by testing if the repeated question has the same answer or not as if the two answers are different, we will delete the response to reduce the biasness.

[7] Phase 5:

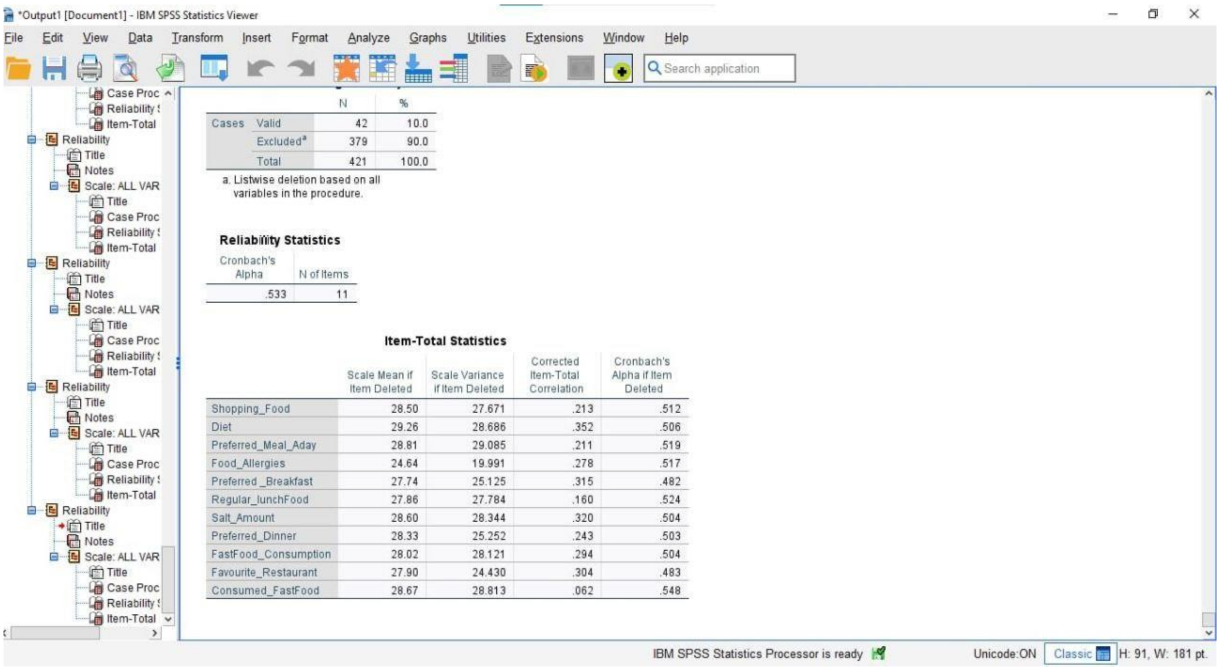
Part 1 :: Expectations VS. Reality: -

- Number of responses that were taken into consideration was 422 as data has been reduced after removing all inconsistent and careless data.
- We were expecting to have about 1,000 responses but since we hadn't had a lot of student to respond to our questionnaire, we couldn't have no more than 527 responses from the target population.

Part 2 :: Questionnaire Validation: -

- After preparing data, we started model validation of the survey to ensure that results would be accurate and reliable.
- We calculated Cronbach's alpha which measures the internal consistency reliability that assesses the degree to which a set of questions are measuring the same construct.

- Cronbach’s alpha value was 0.533 which doesn’t indicate a good internal consistency measure as 0.7 or higher indicates acceptable internal consistency.
- We can say that there is a poor interrelatedness between items. That’s why there is a poor correlation between items so some questions might need to be revised again or to be discarded or we could have gathered more responses so that may be more patterns would reveal.



		N	%
Cases	Valid	42	10.0
	Excluded[a]	379	90.0
	Total	421	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.533	11

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
Shopping_Food	28.50	27.671	.213	.512
Diet	29.26	28.686	.352	.506
Preferred_Meal_Aday	28.81	29.085	.211	.519
Food_Allergies	24.64	19.991	.278	.517
Preferred _Breakfast	27.74	25.125	.315	.482
Regular_lunchFood	27.86	27.784	.160	.524
Salt_Amount	28.60	28.344	.320	.504
Preferred_Dinner	28.33	25.252	.243	.503
FastFood_Consumption	28.02	28.121	.294	.504
Favourite_Restaurant	27.90	24.430	.304	.483
Consumed_FastFood	28.67	28.813	.062	.548

[8] Phase 6: Data Analysis: -

- In this phase, we studied analysis of our data like: descriptive statistics, inferential statistics, correlation analysis, Regression analysis and Data visualization.

- First, we choose some columns to work on it like Favourite_Cuisine, Food Regime, Diet, EatingVarious_Vegetables/Fruits, Shopping_Food, consumed_FastFood and calculate their mean, median, mode and standard deviation then test the significant relationships and differences between variables by chi-square test or t-test.
- We have calculated the mean for each kind of food cuisine to get that nearly 20-23% love Egyptian food, 53% of responses love Chinese food, 3% love Italian food, 3% love Lebanese food and others love all mentioned together.
- Then we assumed median and find it is 7 “Chinese food” that means the middle value is “Chinese food”.
- We also determined the mode and found that it would be 263 “Egyptian food” means that the max number of responses was to “Egyptian food”.
- standard deviation was found to be 71.36 which means that there are variation or dispersion in data in the rate of 71.36.
- At last, we found that there is a relationship between Favourite_Cuisine and (Favourite_Chef, Shopping_Food).

```
> data = read.csv('C:\\Users\\Mega Store\\Documents\\data.csv')
> cuisine = table(data$Favourite_Cuisine)
> mean = cuisine/length(cuisine)
> mean

                                  chinese food                                            Egyptian food
                                    0.53846154                                              20.23076923
                   Egyptian food, chinese food                            Egyptian food, Lebanese food
                                    0.46153846                                               0.61538462
                                  Italian food                             Italian food, chinese food
                                    3.46153846                                               0.23076923
                   Italian food, Egyptian food            Italian food, Egyptian food, chinese food
                                    4.30769231                                               0.69230769
    Italian food, Egyptian food, Lebanese food Italian food, Egyptian food, Lebanese food, chinese food
                                    0.84615385                                               0.23076923
                   Italian food, Lebanese food            Italian food, Lebanese food, chinese food
                                    0.38461538                                               0.07692308
                                 Lebanese food
                                    0.30769231
> median(cuisine)
[1] 7
> mode = max(cuisine)
> mode
[1] 263
> sd(cuisine)
[1] 71.36005
>
> regime = table(data$Food.Regime)
> regime

                                 Meatarian                              None of the above Pescetarian (no meat ,but eat fish and/ or shellfish)
                                       232                                            105                                                    34
                                Vegetarian
                                        50
> mean2 = regime/length(regime)
> mean2

                                 Meatarian                              None of the above Pescetarian (no meat ,but eat fish and/ or shellfish)
                                     58.00                                          26.25                                                  8.50
                                Vegetarian
                                     12.50
> median(regime)
[1] 77.5
> mode2 = max(regime)
> mode2
[1] 232
> sd(regime)
[1] 89.80488
>
> diet = table(data$Diet)
> mean3 = diet/length(diet)
> mean3

   Healthy Non Healthy
      83.5       127.0
```

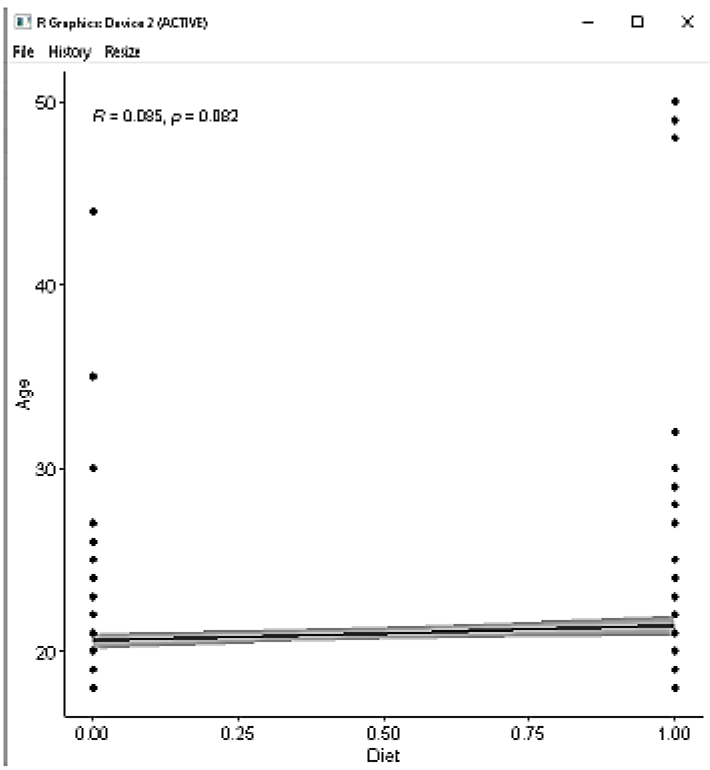
```
> median(diet)
[1] 210.5
> mode3 = max(diet)
> mode3
[1] 254
> sd(diet)
[1] 61.51829
>
> v = table(data$Eatingvarious_vegetables.Fruits)
> mean4 =v/length(v)
> mean4

   NO   Yes
 20.5 190.0
> median(v)
[1] 210.5
> mode4 = max(v)
> mode4
[1] 380
> sd(v)
[1] 239.7092
>
> # p-Value less than the significance level of 0.05, we reject the null hypothesis and conclude that the two variables are in fact dependent.
>
> chisq.test(data$Favourite_Cuisine , data$Food.Regime)

        Pearson's chi-squared test

data:  data$Favourite_Cuisine and data$Food.Regime
X-squared = 35.103, df = 36, p-value = 0.5111

Warning message:
In chisq.test(data$Favourite_Cuisine, data$Food.Regime) :
  Chi-squared approximation may be incorrect
> chisq.test(data$Favourite_Cuisine , data$Favourite_Chef)

        Pearson's Chi-squared test

data:  data$Favourite_Cuisine and data$Favourite_Chef
X-squared = 2605.9, df = 2472, p-value = 0.02999

Warning message:
In chisq.test(data$Favourite_Cuisine, data$Favourite_Chef) :
  Chi-squared approximation may be incorrect
> chisq.test(data$Favourite_Cuisine , data$Shopping_Food)

        Pearson's Chi-squared test

data:  data$Favourite_Cuisine and data$Shopping_Food
X-squared = 459.81, df = 600, p-value = 1

Warning message:
In chisq.test(data$Favourite_Cuisine, data$Shopping_Food) :
  Chi-squared approximation may be incorrect
```

- By applying correlation test, we found that the p-value of the test is 0.02348, which is less than the significance level alpha = 0.05. We can conclude that Diet and Age are significantly correlated with a correlation coefficient of 0.110407 and p-value of 0.02348.

- If correlation coefficient: -
    - -1 indicates a strong negative correlation: this means that every time x increases, y decreases (left panel figure).
    - 0 means that there is no association between the two variables (x and y) (middle panel figure).
    - 1 indicates a strong positive correlation: this means that y increases with x (right panel figure).
- The correlation coefficient in our data between Diet and Age = 0.110407. So, it indicates a weak positive correlation.

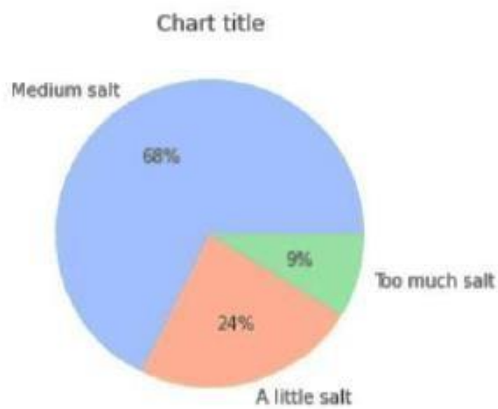
```r
library(ggpubr)
data = read.csv("D:\\university\\Semester4\\Survey\\responses1 (2).csv")
x = data$Diet
y = data$Age

cor_matricx = cor(x,y,method = c("pearson", "kendall", "spearman"))
cor.test(x,y,method = c("pearson", "kendall", "spearman"))

ggscatter(data, x = "Diet", y = "Age",
          add = "reg.line", conf.int = TRUE,
          cor.coef = TRUE, cor.method = "spearman",
          xlab = "Diet", ylab = "Age")

shapiro.test(data$Diet)
shapiro.test(data$Age)

ggqqplot(data$Diet, ylab = "Diet")
ggqqplot(data$Age, ylab = "Age")
```

```
        Pearson's product-moment correlation

data:  x and y
t = 2.2739, df = 419, p-value = 0.02348
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.01499287 0.20382865
sample estimates:
      cor
0.110407
```



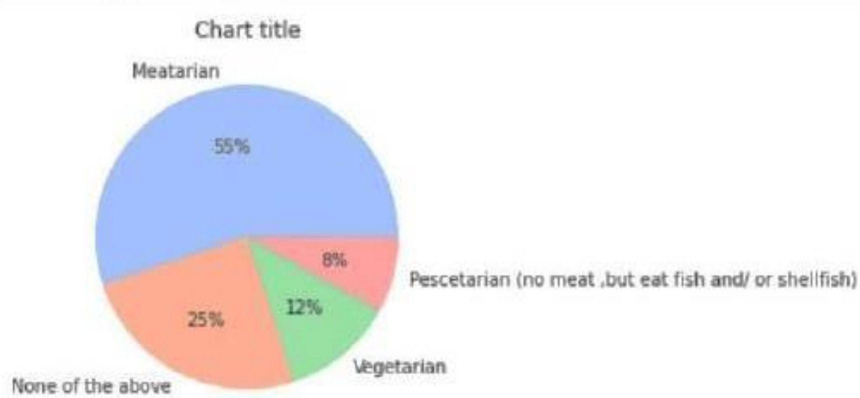
- In the last step in this phase, we started to explore and visualize the data to extract some patterns that would be helpful.

```python
labels = df['Salt_Amount'].value_counts().index
values = df['Salt_Amount'].value_counts().values

plt.pie(values, labels = labels ,colors= colors, autopct = '%.0f%%' )
plt.title('Chart title')

plt.show()
```



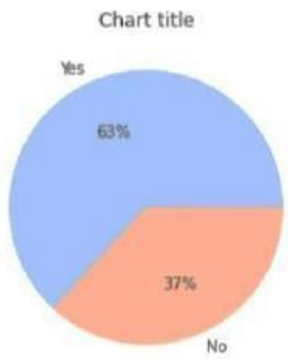
- Most people prefer food with medium amount of salt but some of them would like their food with a little amount of salt.

```python
labels = df['Food Regime'].value_counts().index # Taking the all index
values = df['Food Regime'].value_counts().values  # Taking the all value

colors = sns.color_palette('pastel')

plt.pie(values, labels = labels , colors = colors, autopct = '%.0f%%' )
plt.title('Chart title')

plt.show()
```



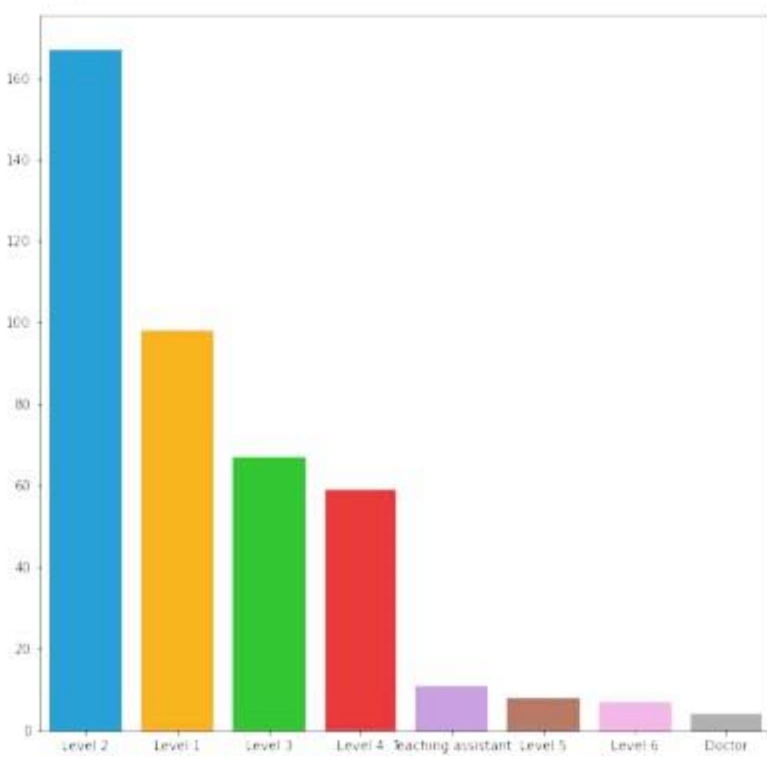
- We can see that nearly 55% of respondents like eating meat rather than vegetables.

```python
labels = df['SpicyFood'].value_counts().index
values = df['SpicyFood'].value_counts().values

plt.pie(values, labels = labels , colors = colors, autopct = '%.0f%%' )
plt.title('Chart title')

plt.show()
```



- We can see that most people like eating spicy food.

```python
In [8]: plt.figure(figsize=(10,10))

        labels = df['Education'].value_counts().index
        values = df['Education'].value_counts().values
        sns.barplot(labels,values)

Out[8]: <AxesSubplot:>
```



-At last, after exploring respondent’s ages we found that most of them are at level 2 and level 1.

[9] Phase 7: Report & Presentation: -

Finally, here comes the last step in the project. A PowerPoint presentation is being prepared to represent our survey project to everybody who is interested to listen. A report is written to tell an overall view about what we did throughout the 7 phases.

[10] Conclusion: -

To sum up, Food preferences survey is a vital tool for collecting data about people’s food choices and habits. These data are essential as we can extract valuable insights that would help take better decisions in the future.